

LAST NAME: _____

Practice Quiz 09C — MATH 2055 — Fall 2026

Practice for Quiz 09

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Auto-generated by the Codex harness using GPT-5.

Write $[x]_8$ for the congruence class of x modulo 8.

1. Consider

$$f : \mathbf{Z}_8 \longrightarrow \mathbf{Z}_8, \quad f([x]_8) = [3x + 1]_8.$$

(a) Prove that f is well defined.

(b) Prove that f is onto and determine f^{-1} .